McCulloch–Pitts Neural Model (MP Neuron)

Theory

- Proposed by **Warren McCulloch** and **Walter Pitts** in **1943**, it is the **first mathematical model of an artificial neuron**.
- It is a **binary threshold model**, meaning the neuron outputs **1 (fire)** or **0 (no fire)** based on a weighted sum of its inputs.
- The neuron simulates the logical behavior of biological neurons using **simple arithmetic and thresholding**.

Structure

- **Inputs ($x_1, x_2, ..., x_n$):** Binary signals (0 or 1).
- **Weights ($w_1, w_2, ..., w_n$):** Strength of each input connection.
- **Summation Unit:** Calculates the total input ($\Sigma x_i w_i$).
- **Threshold (θ):** Fixed limit to decide activation.
- **Activation Function:** $$y = \begin{cases} 1, & \text{if } \sum x_i w_i \geq \theta \\ 0, & \text{otherwise} \end{cases}$$

Logic Gate Simulation

Gate	Weights	Threshold	Logic
AND	$w_1 = 1, w_2 = 1$	$\theta = 2$	Fires only if both inputs = 1
OR	$w_1 = 1, w_2 = 1$	$\theta = 1$	Fires if either input = 1
NOT	$w = -1$	$\theta = 0$	Inverts the input

Characteristics

1. **Binary output:** Only 0 or 1 (no intermediate values).
2. **Fixed weights and threshold:** Learning not included (static model).
3. **Implements Boolean logic:** AND, OR, NOT, NAND, NOR, XOR, etc.
4. **Basis for perceptron:** MP neuron is the foundation of modern neural networks.
5. **No learning mechanism:** Parameters are manually set.

Advantages

- Simple and easy to implement.
- Demonstrates the concept of neuron firing and thresholding.
- Can realize all basic logic gates.

Disadvantages

- Cannot solve **non-linear** problems like XOR directly.
- Only binary inputs/outputs, no real-valued operations.
- No ability to learn or adjust weights automatically.